

ANILA K.

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PROFESSIONAL SUMMARY

AI/ML engineer who ships LLM systems to production and owns them afterwards. Builds multi-agent architectures, retrieval-augmented generation and document understanding pipelines, treating agents as engineered artifacts — versioned, regression-tested and instrumented — rather than prompts edited in a console. Sole AI engineer on an enterprise platform spanning three products, owning system design, model selection, evaluation and the backend and integration work around them. MSc in Intelligent Systems, with prior work across NLP, computer vision and deep learning.

PROFESSIONAL EXPERIENCE

Software Engineer (AI/ML) | Sybrant Technologies, Chennai

Aug 2025 - Present

Sole AI engineer on PactPro, an enterprise document-intelligence platform spanning three products — Abstraction (AI document review), Draft (AI document generation) and AMS (workflow management) — plus LeaseRush. Owned model selection, retrieval architecture, prompt engineering, evaluation and the backend systems around them. Domain: legal.

Retrieval-Augmented Generation & Document Understanding

- Designed and shipped an end-to-end contract abstraction pipeline: Docling PDF parsing with hybrid chunking, **nomic-embed-text** embeddings in PostgreSQL/**pgvector** (migrated from FAISS for transactional consistency and multi-tenant isolation), and **Qwen 2.5 32B** served locally via Ollama behind a LiteLLM proxy with Anthropic API fallback, automating review across lease, MSA, NDA, loan, shareholders, licence and underwriting agreements.
- Self-hosted quantized open-weight inference on local hardware so privileged client documents never leave client infrastructure — sizing the model to available CPU/GPU, tuning throughput against latency targets, and removing per-token API cost from the primary extraction workload entirely.
- Engineered a production prompt library spanning leases, MSAs, NDAs, loan agreements, shareholders agreements (**57-clause taxonomy**), licence agreements and SEBI ICDR underwriting agreements, evaluated against held-out document sets and regression fixtures so accuracy on difficult clause families — force majeure, holdover, CAM/OpEx, survival, assignment, authority to sign — was measured rather than assumed, and drift surfaced in the evaluation rather than in client escalations.
- Established retrieval-vs-generation separation as a core design principle — keyword-dense retrieval queries tuned independently of extraction prompts, plus false-positive guards and decoy exclusions — mapping failure modes explicitly rather than discovering them in production. Diagnosed and rebuilt match vocabularies for Commonwealth/Indian drafting conventions, where US-centric terminology caused systematic recall loss that no amount of prompt tuning would have surfaced.

Agentic AI — Multi-Agent LLM Pipeline for Document Generation

- Architected a **six-agent LLM drafting pipeline** (Intent, Clause Locator, Rewriter, Consistency, Risk Review, DOCX Generation) over a five-layer system design — document-type agnostic, so firms draft contracts, agreements, notices, petitions and bespoke instruments through the same pipeline rather than fixed templates. Produces a reviewable first draft in a single pass, replacing manual assembly from precedent files.
- Designed a block-based document model with stable block IDs and a **propose-review-commit** pattern, so every agent action is attributable, reviewable and reversible rather than an opaque rewrite — a hard requirement for professional adoption.
- Treated prompts and agent behaviour as **versioned artifacts**, stored as structured database records against document type and clause rather than inline strings, so a change is reviewable and traceable instead of depending on who last edited a prompt.
- Built automated **playbook generation** from a firm's own precedent documents, defining a Preferred / Fallback / Red-line / Rationale schema so each firm's drafting standards are learned from its corpus and used to risk-score generated clauses — no hand-authored rules per document type. Benchmarked positioning against Harvey AI, SpotDraft and Avvoka.

Full-Stack Platform Engineering — Advocate Management System

- Delivered a production practice-management platform for law firms on **Java 21 / Spring Boot 3.5 / React 19 / MySQL** — 24 REST controllers, 17 JPA entities and 25 API modules covering clients, cases, hearings, documents, invoicing, expenses, analytics and reporting.
- Implemented **role-based access control** with 6 roles and 36 granular permissions across 8 modules, enforced declaratively via a Spring AOP aspect, plus a forensic audit trail capturing actor, IP, device and request context for compliance.
- Built a multi-channel notification engine (SMTP, WhatsApp via Meta/Twilio behind a provider-strategy interface, in-app STOMP WebSocket alerts) with an async queue handling retries, rate limiting and duplicate protection.
- Hardened the platform for production: JWT auth, BCrypt hashing, hashed OTP reset with rate limiting, magic-byte file-upload validation, security-header policy, and performance tuning via pagination, **@EntityGraph** N+1 elimination and composite indexing.
- Shipped an **AI assistant** exposing case and client data through a natural-language query interface, alongside OpenPDF-generated reports, invoices and receipts across 10+ document types and a multi-type backup/restore system.

Data Acquisition, Integration & Workflow Automation

- Built a **FastAPI scraper microservice** for Indian court portals (Madras High Court case status and live display board, eCourts District & High Court Services, Supreme Court of India) on a **CNR-keyed data model** — resolving case type/number/year to CNR once at onboarding, then polling cheaply thereafter, covering the three-tier Tamil Nadu judicial hierarchy from district courts through the High Court and Supreme Court.
- Integrated against systems never designed to be integrated with, engineering for their failure modes: a JWT-gated gateway rather than browser exposure, tiered caching (24h registry, 1h results, fetch-once persistence for ~352 static case types), bounded retries with exponential backoff on 502/503/504, terminal handling of 4xx, and deliberate serialisation with no parallel fan-out to stay within what the upstream could bear.
- Modelled the eCourts district-court **stateful cascade** (state to district to complex to establishment to case type) generically in the proxy layer with no per-step code, mapping scraped records into a guided intake flow with prefilled fields.
- Integrated AMS, Abstraction and Draft into one platform: consolidated identity onto SimpleJWT, planned a BCrypt credential migration off the legacy Spring Boot service, and remediated hardcoded API keys and CORS issues.
- Shipped a statutory reference migration feature with synthetic PDF fixtures; evaluated Indic translation models (IndicTrans2, Sarvam-Translate, Krutrim-2) for document-level translation with domain glossaries; automated recurring briefing workflows with **n8n** agents over Gmail and Calendar.

TECHNICAL SKILLS

Agentic AI & Evaluation: Multi-agent system design, tool integration & orchestration, context engineering, structured outputs, schema-driven extraction, eval harness design, held-out & regression testing, failure-mode analysis, accuracy/latency/cost tuning under production load

LLM & Agents: Multi-agent pipeline design, context engineering, tool integration, structured outputs, evaluation suites & regression fixtures, versioned prompt artifacts, RAG architecture, embeddings & vector search (pgvector, FAISS), small language models (SLMs) & open-weight models (Qwen 2.5), parameter-efficient fine-tuning (LoRA, QLoRA, PEFT), model quantization & inference optimisation, on-premise/self-hosted serving (Ollama), inference latency & CPU/GPU utilisation tuning, model routing & fallback (LiteLLM), Anthropic & OpenAI APIs, LangChain, Hugging Face Transformers & PEFT (PyTorch)

Machine Learning: Supervised & deep learning, transfer learning and model fine-tuning (YOLOv8), NLP (BERT, SentenceTransformers, NLTK), computer vision (OpenCV), time-series forecasting (LSTM), PyTorch, TensorFlow, Keras, scikit-learn, benchmarking & held-out evaluation, error analysis, data annotation & dataset curation

Production Engineering: End-to-end ownership of shipped systems, failure-mode design, evaluation-driven iteration, latency/cost/reliability tuning, on-premise & gateway model serving with fallback routing, monitoring of latency and resource utilisation, retry & backoff strategy, tiered caching, Git, Linux

Backend: Python, Java 21, Django, Django REST Framework, FastAPI, Spring Boot 3, Spring Data JPA/Hibernate, Spring AOP, Celery, Redis, PostgreSQL, MySQL, Flask, REST API & microservice design

Document AI / IDP: Intelligent Document Processing, structured extraction from unstructured documents, Docling, PyMuPDF, ReportLab, python-docx, OCR & layout-aware parsing, chunking strategies, schema-driven field extraction

Data & Tooling: pandas, NumPy, web scraping (requests, BeautifulSoup, Selenium), Tesseract OCR, n8n workflow automation, Git, Linux, REST APIs

Real-time & Security: WebSocket (STOMP/SockJS, Django Channels), JWT authentication, RBAC with AOP enforcement, BCrypt, secure file upload validation, audit logging, caching & retry strategy

Frontend: React 19, TypeScript, Vite, TipTap, Recharts, Streamlit, Gradio

Domain Exposure: Legal & contract document AI, regulated-industry document workflows, data-residency-constrained deployment, e-commerce policy QA, clinical-video computer vision, financial time-series

PROJECTS

SLM Fine-Tuning & Quantization LoRA, QLoRA, PEFT, Hugging Face	MSc Research
• Fine-tuned small language models using parameter-efficient adapter methods (LoRA, QLoRA) with Hugging Face and PEFT, comparing adapter rank and precision settings against full fine-tuning on constrained hardware. • Applied quantization to reduce model footprint and inference latency, benchmarking accuracy retention against memory and CPU/GPU utilisation to identify viable deployment configurations.	
RAG Legal & T&C Assistant for Indian E-Commerce Platforms LLM, RAG, FAISS	Jul 2025
• Built a RAG system over legal PDFs and platform policies of Amazon, Flipkart, Meesho and Ajio using FAISS retrieval with Falcon-7B generation; SentenceTransformers embeddings, LangChain chunking, Gradio interface deployed on Hugging Face Spaces.	

EDUCATION

MSc Computer Science (Intelligent Systems), GPA 8.15/10 Central University of Kerala	2022 - 2024
BSc Physics, GPA 8.83/10 University of Calicut	2019 - 2022
Languages: English (Proficient) · Malayalam (Native) · Hindi (Conversational) · Tamil (Conversational)	